

# Rishi More

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## Summary

**ML engineer** focused on **efficient and reliable large language model (LLM) inference**: CUDA/kernel profiling, fused quantized kernels, and **vLLM**-based calibration pipelines. **Co-first author, ICML 2026 Main Conference**, on distribution-free risk control for cutting test-time reasoning compute under calibrated error guarantees.

## Experience

### JHU Data Science and AI Institute (DSAI)

May 2025 – Present

*Graduate Research Assistant, LLM Inference Optimization*

*Baltimore, Maryland*

- **Co-first author** of *Conformal Thinking* (**ICML 2026**, full paper): finite-sample calibration that halts reasoning early while holding a target error rate.
- Built a vLLM-based calibration/evaluation pipeline across **4** reasoning models and **4** benchmarks (**2.7K+** problems), maintained target error rates under dataset/length shift using finite-sample (UCB) calibration.
- Designed dual-threshold stopping rules, including a parametric lower threshold and ensembled halt policy, reducing reasoning tokens by **12%** at matched accuracy with reusable evaluation tooling.

### JHU Center for Language and Speech Processing (CLSP)

January 2025 – December 2025

*Graduate Researcher, Retrieval Systems*

*Baltimore, Maryland*

- Built a temporal RAG pipeline over **10K+** time-stamped documents with time-decay weighting and metadata-scoped indexing to prioritize recent evidence over stale matches.
- Raised retrieval precision **23%** over a dense-embedding baseline by down-weighting outdated passages in fast-changing topics.
- Reduced stale and hallucination-prone retrievals **31%** via reference-free consistency scoring on retrieved context.

### Mastek

November 2022 – July 2023

*Machine Learning Engineer Intern*

*Mumbai, India*

- Built and deployed an on-prem **edge** voice authentication system at **<100ms p99** latency and **96.6%** accuracy, using a lightweight Siamese mel-spectrogram model for retraining-free enrollment under edge-compute limits.
- Engineered a TensorFlow pipeline over **1M+** voice samples with spectral-gating denoising for robust feature extraction.
- Improved authentication reliability **20%** in high-noise production environments via preprocessing and feature-extraction upgrades.

## Projects

### Post-Training Evaluation & Reward-Hacking Detector | *Python, PyTorch, DPO, LLM-as-Judge* | GitHub

- Built an evaluation harness for **6** post-training failure modes—reward hacking, sycophancy, verbosity bias, false confidence, safety drift, and distribution shift—across **4** Qwen3-8B variants: Base, supervised fine-tuning (SFT), Direct Preference Optimization (DPO), and Instruct.
- Evaluated **142** adversarial prompts with debiased pairwise LLM judging, position-swapped comparisons, and log-prob calibration, flagging **10%** of DPO preference wins that degraded factuality.
- Diagnosed optimization-stage regressions using ECE, Brier score, AUROC, and bootstrap confidence intervals, finding SFT raised harmful compliance to **12.5%** and DPO increased sycophancy from **34%** to **45%**.

### Scaffolded Multi-Agent Training for Language Agents | *Python, LLM Agents, LoRA, asyncio, BM25* | GitHub

- Built an async multi-agent training system (Tinker API): a frozen **Qwen3-235B** caregiver teaching Low-Rank Adaptation (LoRA)-tuned **Qwen3-8B** agents via four-module memory and salience-gated BM25 retrieval.
- Achieved **100%** training success with **25%** fewer turns (6.1 vs. 8.4) across **4** conditions and **3** seeds versus Solo/Peer baselines.
- Isolated scaffolding dependency as the transfer bottleneck via ablations; packaged experiments behind a fully reproducible single-command CLI.

### Decode Roofline: Kernel Profiling & Fused INT4 Dequant-GEMV | *CUDA C++, Nsight, vLLM* | GitHub

- Profiled batch-1 LLM decode (**Qwen2.5-1.5B**) at CUDA-kernel level, identifying **~81%** of kernel time in weight-streaming GEMVs pinned at **~95%** of the measured **~250 GB/s** memory roofline.
- Built correctness-gated fused **INT4** dequant + GEMV CUDA kernel hitting **86%** of peak bandwidth, reduced memory traffic by **~8.7×** versus a two-op baseline and improved batch-1 latency by **2.38×**.

## Education

### Johns Hopkins University

May 2026

*M.S.E. in Computer Science*

*GPA: 3.9/4.0*

- **Coursework:** Theory of Replicable ML, Human-in-the-Loop ML, Non-Stationary Environments.

### University of Mumbai

July 2024

*B.Tech. in Computer Science, Honors in AI/ML*

*CGPA: 9.85/10*

## Technical Skills

**Languages:** Python, C++, Java, C

**ML/LLM Systems:** PyTorch, TensorFlow, Hugging Face, Low-Rank Adaptation (LoRA), Direct Preference Optimization (DPO), retrieval-augmented generation (RAG), calibration, conformal/risk control, Scikit-learn

**Inference & MLOps:** vLLM, CUDA C++, Nsight Compute/Systems, Docker, Kubernetes, AWS, Apache Airflow, Apache Spark

**Data:** SQL, PostgreSQL, MongoDB, Pandas, NumPy